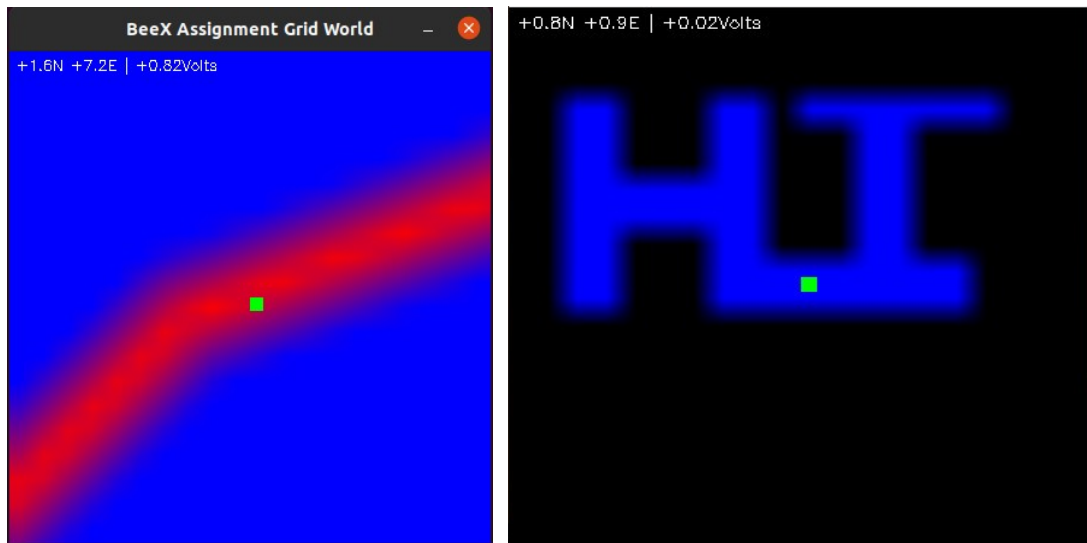


Task 1: Mission Planning for pipeline inspection

This task is to develop a simple Autonomous Subsea Pipeline Tracking system for a simulated Hovering Autonomous Underwater Vehicle (HAUV) operating in a low-visibility environment. The core task involves creating two interconnected **ROS1** packages:

1. a *driver* package responsible for low-level communication (TCP/UDP/Serial) with the simulator to handle movement commands, odometry data, and Field Gradient (FG) sensor readings;
2. a *mission* package that implements the high-level control logic. The driver must abstract simulator protocols into standard ROS topics (`/odom`, `/fg_readings`) and services (`/command`).

The mission objective is to locate a North-South oriented pipeline and autonomously track the centerline from its southern to northern terminus. Success is defined by first locating the southern most start point of the pipeline and then tracking (within 0.5m lateral distance) the pipeline to the most northern end, adhering to a 100m x 100m operational boundary. This assignment evaluates proficiency in C++17, ROS 1 modular design, network fundamentals (TCP/UDP/Serial), and the development of mission planning strategy for sensor-based navigation. See the README in the task documents for more details.



Examples of the grid world game

In your submission for this task, please submit your C++17 code and a README file that includes 1) A concise explanation of your mission strategy, 2) Instructions to compile and run your solution. Your solution will be evaluated on Ubuntu 20.04 with ROS1 Noetic.